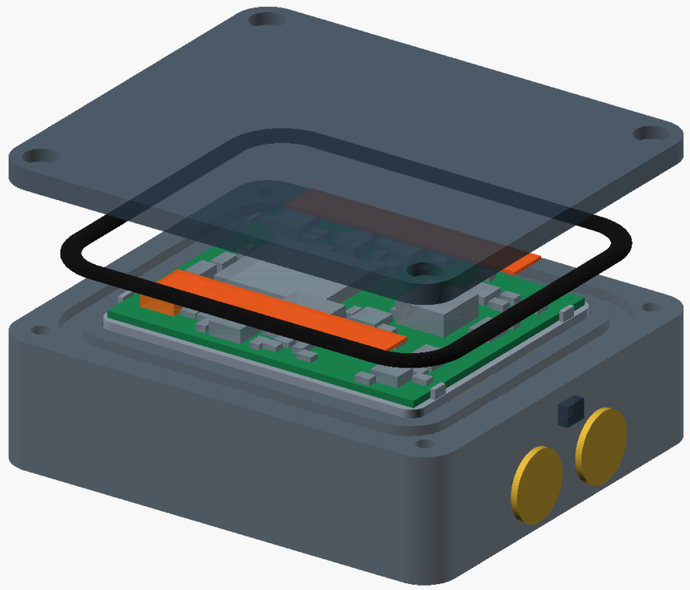
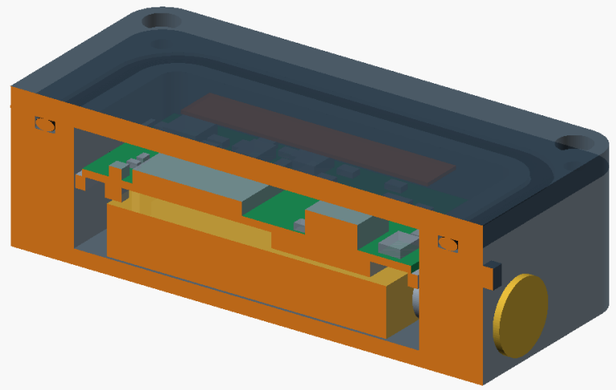
COLLAR TRACKER / r07 ENCLOSURE REVIEW

Actual PCB placements and carrier clearance. Pack and antenna boxes remain reserved volumes.

EXPLODED / shell, seal and populated envelopes



SECTION / internal clearance model



VERIFIED CAD GEOMETRY

85 populated envelopes; four PCB thickness cases pass.

New U12 relief and one moved support leg; 1 mm bridge.

Actual CAD clearance intersection is empty.

Previous carrier control has 1.234 mm³ interference.

Eight exported parts: closed, oriented and connected.

Rigid body + lid: 57.4 × 49.4 × 17.6 mm (key excluded).

Boot envelope: 60.4 × 52.4 × 22.4 mm.

WATER / DUST TARGET — UNQUALIFIED

IP68 swimming/dust target; no physical test yet.

Provisional immersion: 2 m fresh water / 60 minutes.

Pack and antenna dimensions still need real selections.

Material/seal grades and fastener torque remain open.

Harness, dock wiring and positive PCB retention: open.

Carrier strength and thermal coupling require tests.

Grey/orange blocks are fit bounds, not qualified parts.

Model: mechanical/swim-r07-charge-interlock; exact input hashes in mesh-checks.json.

PCB and enclosure evidence are separate from assembled electrical and ingress acceptance.